

PRODUCT DATA SHEET

VIR KLADS (EDS / EDF)



DESCRIPTION

VIR KLADS is an Exterior grade compact laminate designed for durable and aesthetic cladding in commercial and residential facades. Available in both fire-grade and general-purpose options, it features décor paper with lightfastness >5 and a UV-protection film over 50 µm for enhanced weather resistance. Engineered to withstand outdoor weathering and varied climatic conditions, VIR KLADS ensures long-lasting performance, stability, and visual appeal for exterior applications.

Description: VIR KLADS

Laminate Grade 1: EDS (Exterior Grade Standard)

Laminate Grade 2: EDF (Exterior Grade Fire retardant)

Sizes: 1220x2440 mm, 1220x2800 mm, 1220x3050 mm, 1320x2800 mm, 1320x3050 mm, 1320x4200 mm, 1830x3660 mm, 1830x4330 mm

Thickness: 6 mm – 12 mm

Core Construction: Phenolic Kraft paper core available in standard brown or black core configuration, depending on product specification and décor requirements.

Note: All technical properties and performance values stated in this document are based on testing conducted on 6 mm VIR KLADS exterior compact laminate panels unless otherwise specified

APPLICATION / APPLICATION AREA

Ideal for exterior cladding, ventilated facades, balcony paneling, soffits, and other outdoor architectural surfaces.



FIRE PERFORMANCE (ON DEMAND)

VIR KLADS (EDF Grade) is CE certified for exterior applications and evaluated for fire performance in accordance with EN 13501-1:2018, ASTM E84-23, BS 476 Part 6 & Part 7, and UL 94:1997. Fire testing has been conducted on 6 mm thickness exterior compact laminate panels to evaluate reaction to fire, surface burning characteristics, smoke development, fire propagation, and flammability performance under controlled laboratory conditions.

Test / Parameter	Test Standard	Requirement / Criteria	Observed Results	Classification / Status	Conformity
A. Single Burning Item (SBI) Test					
LFS to Edge	EN 13501-1:2018	No LFS to edge	No	B	Confirms
FIGRA ₀₋₂ MJ (W/s)	EN 13823:2020 + A1:2022	≤ 120 W/s	16.15		Confirms
THR _{600S} (MJ)		≤ 7.5 MJ	4.42		Confirms
SMOGRA (m²/s²)		≤ 30 m²/s²	0	s1	Confirms
TSP _{600S} (m²)		≤ 50 m²	29.11		Confirms
Flaming Droplets / Particles ≥ 10 sec within first 600 sec		No	No	d0	Confirms
B. Ignitability Test					
Flame to Reach 150 mm Mark	EN ISO 11925-2	Flame should not reach 150 mm mark	No flame reached 150 mm mark	Pass	Confirms
C. Surface Burning Characteristics					
Flame Spread Index (FSI)	ASTM E84-23	0 – 25	20	Class A / Class I	Confirms
Smoke Developed Index (SDI)		0 – 450	80	Class A / Class I	Confirms
D. Surface spread of flame					
Surface Spread of Flame	BS 476 Part 7:1997	Class 1 Limits	Class 1 Achieved	Class 1	Confirms
Flame Spread at First 1½ Minutes		≤ 165 mm	0 mm	Class 1	Confirms
Final Flame Spread		≤ 165 mm	0 mm	Class 1	Confirms
E. Fire Propagation Test					
Fire Propagation Index (I)	BS 476 Part 6:1989 + A1:2009	Tested Value	11.6	Reported	Confirms
Sub-index i ₁		–	2.3	Reported	Confirms
Sub-index i ₂		–	3.5	Reported	Confirms
Sub-index i ₃		–	5.8	Reported	Confirms
F. Vertical Burning Test					
After Flame Time after First Flame Application (t ₁)	UL 94:1997 – 20 mm	< 10 sec (V-0)	0 sec	V-0	Confirms
After Flame Time after Second Flame Application (t ₂)		< 10 sec (V-0)	0 sec	V-0	Confirms
Total After Flame Time (t ₁ +t ₂) for 5 Specimens		< 50 sec (V-0)	0 sec	V-0	Confirms
After Glow Time after Second Flame Application (t ₃)		No requirement	0 sec	V-0	Confirms
After Flame + After Glow Time (t ₂ +t ₃)		< 30 sec (V-0)	0 sec	V-0	Confirms
Flame or Glow Reaching Holding Clamp		No	No	V-0	Confirms
Ignition of Cotton by Flaming Particles / Drops		No	No	V-0	Confirms
Overall UL 94 Classification		V-0 Requirements	V-0 Achieved	UL 94 V-0	Confirms

QUALITY CHARACTERISTICS / TECHNICAL DATA

KLADS of both General purpose and Flame- retardant (EDS/EDF) are classified under EN-438-6 and tested in accordance to EN-438-2 to verify their quality, durability, and safety. The technical performance values and test results presented are based on testing conducted on 6 mm thickness panels.

Properties	Testing method	Assessment	Standard value ¹⁾	Typical value ²⁾
Weatherability properties				
Resistance to artificial weathering (3000 hours)	EN 438-2:2016, Section 29	Contrast: Gray scale Appearance: Degreeeqq	≥ 3 4	4-5 5
Resistance to UV rays (1500 hours)	EN 438-2:2016, Section 28	Contrast: Gray scale Appearance: Degree	≥ 3 ≥ 4	4-5 5
Resistant to Climate Shock	EN 438-2: 2016, Section 19	Appearance: Degree Flexural Strength Δ_s Flexural Modulus Δ_m	≥ 4 ≥ 0.95 ≥ 0.95	5 0.98 0.95

Properties	Testing method	Assessment	Standard value ¹⁾	Typical value ²⁾
Mechanical properties				
Bulk density	EN 323	g/cm ³	≥ 1.35	1.47
Bending strength: Parallel to long side of panel	EN ISO 178	Mpa	≥ 80	136.1
Bending strength: Parallel to short side of panel	EN ISO 178	Mpa	≥ 80	98.8
Modulus of elasticity: Parallel to long side of panel	EN ISO 178	Mpa	≥ 9,000	15336
Modulus of elasticity: Parallel to short side of panel	EN ISO 178	Mpa	≥ 9,000	11088
Tensile Strength	EN ISO 527-2	Mpa	≥ 60	120
Resistance to impact with a large diameter ball	EN 438-2:2016, Section 21	mm mm	Drop height ≥1400 Impact Outcome <10	1800 5
Light fastness	EN 438-2:2016, Section 27	Grey Scale Rating	4-5	5
Resistance to staining	EN 438-2:2016, Section 26	Group 1 & 2 Group 3	= Grade 5 ≥ Grade 4	5 5
Resistance to scratching	EN 438-2:2016, Section 25	N	≥ 3 ≥ Grade 3	6 4
Resistance to crazing	EN 438-2:2016, Section 24	Rating	≥ Grade 4	5
Screw Pull out strength	ISO 13894-1: 2	N	-	3514

Thermal properties				
Coefficient of Linear Thermal Expansion	ISO 11359-2	mm/mm °C	-	6.81x10 ⁻⁶
Thermal Conductivity	EN 12524	W/mK	≤ 0.30	0.2
Dimensional stability at elevated temperature	EN 438-2:2016, Section 17	longitudinal: % transversal: %	≤ 0.30 ≤ 0.60	0.10 0.08
Resistance to dry heat (160 °C)	EN 438-2:2016, Section 16	Appearance	≥ Grade 4	5
Resistance to immersion in boiling water	EN 438-2:2016, Section 12	mass increase: % (max) thickness increase: % (max) Appearance	2.0 2.0	0.4 0.2 5
Resistance to water vapour	EN 438-2:2016, Section 14	mass increase: % (max) Appearance	5.0 ≥ Grade 4	1.0 5
Resistance to surface wear	EN 438-2:2016, Section 10	Revolutions	≥ 150	450
Resistance to wet heat (100 °C)	EN 438-2:2016, Section 18	Appearance : Gloss Finish	≥ Grade 4	5

SURFACE FINISH

The surface finish test is conducted in accordance with EN 13722 to evaluate the angle of light reflection for each surface finish. The results are expressed in gloss units (GU), providing a precise measure of the reflective characteristics.

SURFACE	MAXIMUM DEVIATION	GLOSS UNITS
High Gloss Surface	± 15	> 70
Semi-Gloss Surface	± 10	30 - 70
Suede Surface	± 5	10 - 30
Matt surface	± 3	< 10

DIMENSIONAL TOLERANCES

NOMINAL THICKNESS (mm)	THICKNESS TOLERANCE (mm)	LENGTH & WIDTH (mm)	STRAIGHTNESS OF THE EDGES (mm/m)	SQUARENESS (mm/m)	FLATNESS (mm/m)
2-3	±0.2	+10/-0	≤ 2	≤ 1.5	≤ 2
3-5	±0.3	+10/-0	≤ 2	≤ 1.5	≤ 2
5-8	±0.4	+10/-0	≤ 2	≤ 1.5	≤ 2
8-12	±0.5	+10/-0	≤ 2	≤ 1.5	≤ 2
12-16	±0.6	+10/-0	≤ 2	≤ 1.5	≤ 2
16-20	±0.7	+10/-0	≤ 2	≤ 1.5	≤ 2
20-25	±0.8	+10/-0	≤ 2	≤ 1.5	≤ 2

APPLICATION

UV protection is available in both single-sided (1 UV) and double-sided (2 UV) configurations, selected based on application exposure conditions and customer requirements.

Grade	Application
EGS 1 UV	Exterior Grade Standard laminate with UV protective film on one side. Used where only one face of the Compact laminate is exposed to direct sunlight, such as building facades and wall cladding.
EGS 2 UV	Exterior Grade Standard laminate with UV protective film on both sides. Used for balcony paneling, parapets, louvers, and areas where both faces are exposed to sunlight and weathering.
EGF 1 UV	Exterior Grade Fire Retardant laminate with UV protective film on one side. Used where fire performance is required and only one face is exposed to direct sunlight, such as facades, external wall cladding.
EGF 2 UV	Exterior Grade Fire Retardant laminate with UV protective film on both sides. Used in high-risk exterior applications where both faces are exposed to sunlight and weathering and fire safety compliance is mandatory, such as balconies, and public building exteriors.